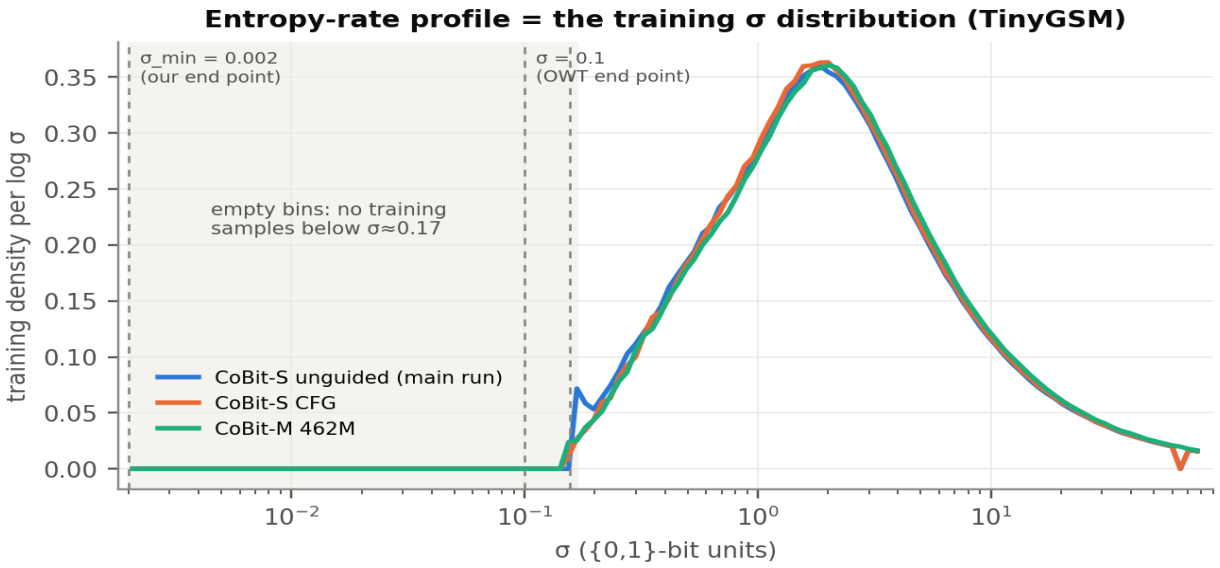
Entropy-rate profile and where the sampler should stop

TinyGSM CoBit runs · the stored entropy tables, the training MSE they imply, and the step grid · 13 Oct 2026 · CPU only

The model is essentially never trained below σ ≈ 0.17, yet we sample down to σ = 0.002. 54 of the 128 bins of the learned training distribution are empty, all at low σ. After the 40k-step warm-up, training draws σ from this profile, so the network sees almost no σ below ≈0.17 for the remaining ~460k steps (and the emptiness is self-sustaining: the controller only measures error where it samples). The sampler's last step then jumps from σ = 0.157 to 0.002 in one go (gain 0.99) into that region. Ending at σ = 0.1, as on OWT, keeps the trajectory where the model was trained and cuts the last step's gain to 0.36.

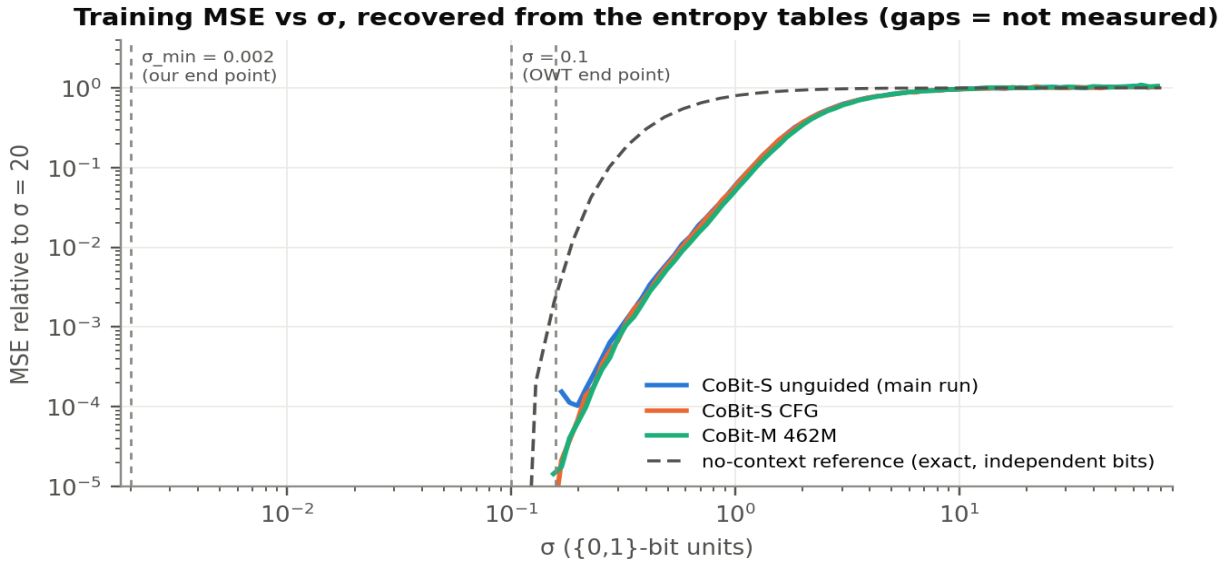
Our σ_min is 0.002, not 2e-4 (configs/tasks/tinygsm_bits.py, EDM's default, in {0,1}-bit units; 0.004 in ±1 units). Below σ ≈ 0.13 the matched-filter logit $(x - ½)/σ^2$ exceeds its clip of ±30, so the prediction there is close to the analytic hard decision, which is why this has not produced visible numerical blow-ups (and the drift $(x - D)/σ$ stays O(1) in fp32). The risk is the learned logit, which is evaluated there almost untrained.

1 The entropy-rate profile



Stored training σ distribution of each run (128 log-spaced bins on [0.002, 80]). Dashed lines: our end point 0.002, the OWT end point 0.1, and 0.157, the last point the entropic grid places before jumping to σ_min. All three runs agree; the peak is at σ ≈ 1.9–2.0.

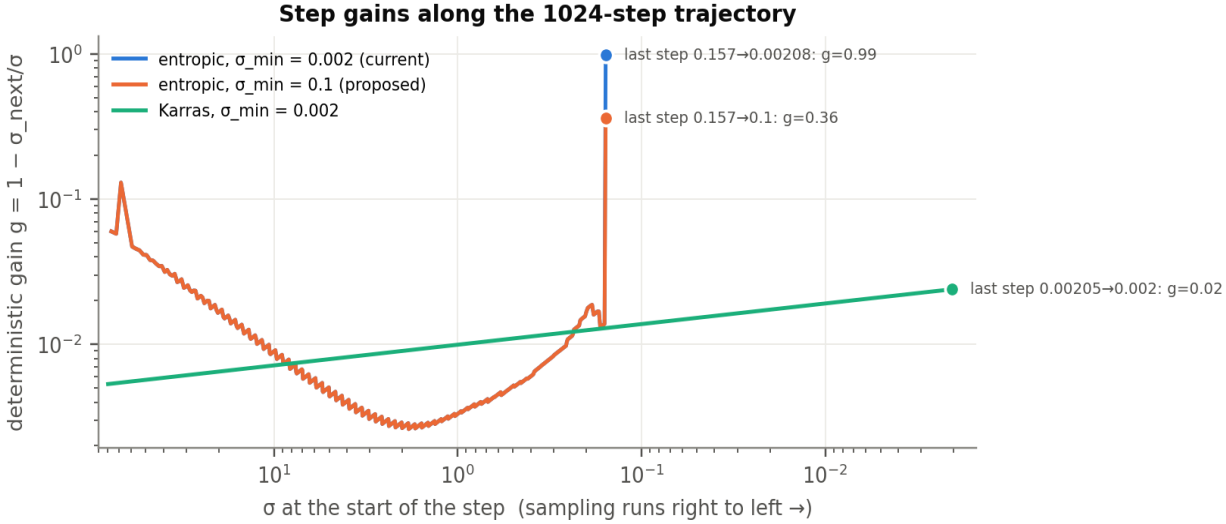
2 MSE vs σ



Training MSE per bin implied by the tables, MSE ∝ σ² (pdf/g)², scaled to 1 at σ=20; dashed: the exact MMSE of an isolated bit with no context, scaled to its prior variance. The gap between them is what context buys. Empty bins are left blank: they were not measured, which is not the same as zero error.

- Relative MSE is 8.7e-04 at σ = 0.3, 0.067 at σ = 1, 0.63 at σ = 3 and ≈1 above σ ≈ 10. Even an isolated bit, with no context at all, is decided from x alone by σ ≈ 0.1 (relative MMSE 6e-09; dashed line), so nothing is gained by integrating below it.
- The informative band — where the model's error falls fastest and the schedule spends its steps — is σ ≈ 0.3–10. Nothing in the profile argues for steps below σ ≈ 0.1.

3 The step grid, and the proposed end point



Per-step deterministic gain along the 1024-step grids (EDM churn adds ≈ γ/(1+γ) to every step). The current entropic grid has one isolated step of gain 0.99 at the very end; ending at σ = 0.1 removes it.

end point σ_min	last step	deterministic gain	with churn γ=0.41	no-overshoot needs τ ≥	stable needs τ >
0.002 (current)	0.157 → 0.002	0.99	0.99	0.99	0.50
0.05	0.157 → 0.05	0.68	0.77	0.77	0.39
0.1 (as on OWT)	0.157 → 0.1	0.36	0.55	0.55	0.27

Ending at 0.1 does not make a sharpened last step safe on its own with churn (it still overshoots below τ ≈ 0.55), so it combines naturally with leaving the last step unsharpened; it does remove the untrained-σ exposure and the one-step jump. Readout at σ = 0.1 is effectively analytic: the matched-filter term is already at its clip there.

4 Updated run plan (nothing submitted)

block	what	cells	GH200-h
A	unsharpened last step; churn γ=0.41 and γ=0; τ ∈ {0.5, 0.3, 0.2}; trajectory diagnostics	6	1.6
B	ancestral sampler, unsharpened last step, τ ∈ {1, 0.7, 0.5, 0.3, 0.2, 0.1}	6	1.6
C	γ ∈ {0.1, 0.2, 0.3} × τ ∈ {0.5, 0.3, 0.2, 0.1}, unsharpened last step	12	3.2
D	DeepMind logit temperature on CoBit, T ∈ {0.5, 0.3, 0.1}	3	≥ 0.8
E (new)	end at σ_min = 0.1, churn γ=0.41: τ ∈ {1, 0.7, 0.5, 0.3, 0.2} with the last step sharpened, plus τ ∈ {0.3, 0.2} unsharpened	7	1.9

All on the 500k EMA checkpoint, entropic grid, 1024 steps, 1319 problems; ≈ 9–10 GH200-h in total, ≈ 20 min wall per block. If E confirms the end point, the cleanest follow-up is to move A–C to σ_min = 0.1 rather than add more cells. Data: the entropy tables of the three runs; the step grids from SigmaSchedule. Built by docs/pdf_reports/build_entropy_profile.py.